Similar Triangles

Name _____ Date _____ Calculator allowed Version 1

Scan to practise this topic online

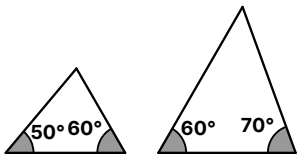
KEY WORDS similar *(rite)* · scale factor *(tauwehe tauine)* · corresponding sides · enlargement *(whakarahi)*

A Are they similar? ●○○

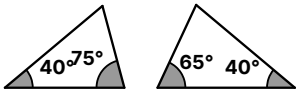
Remember: Two triangles are similar when they have the same three angles. Two matching angles are enough: the third angle is then the same too.

1. Decide whether each pair of triangles is similar. Give a reason.

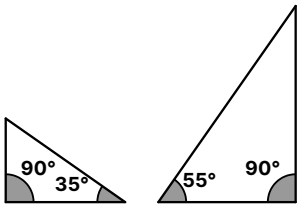
a)



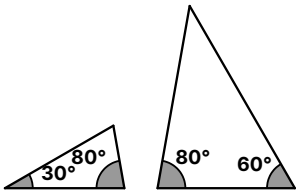
b)



c)



d)



2. These are the side lengths of two triangles. Are the triangles similar?

a) 3, 4, 5 cm and 7.5, 10, 12.5 cm

b) 4, 6, 8 cm and 6, 9, 13 cm

c) 6, 8, 9 mm and 9, 12, 13.5 mm

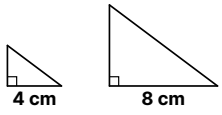
d) 5, 7, 10 m and 12, 17, 24 m

B Scale factor ●○○

Remember: Scale factor k = a side of the image ÷ the matching side of the original. Match the sides that are in the same position.

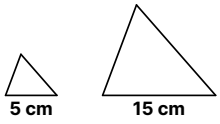
3. Each pair of triangles is similar. Find the scale factor k from the small triangle to the large one.

a)



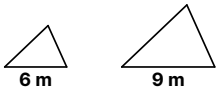
$k =$ _____

b)



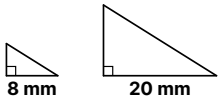
$k =$ _____

c)



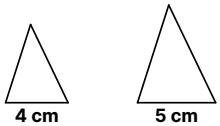
$k =$ _____

d)



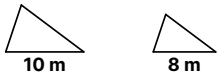
$k =$ _____

e)



$k =$ _____

f)



$k =$ _____

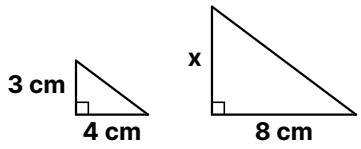
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C Finding a missing length ●●○

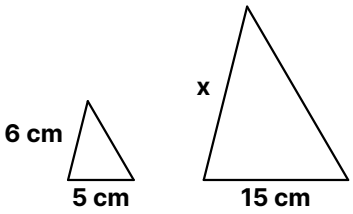
Remember: Find k from a pair of sides you know. Multiply by k to go to the large triangle, divide by k to come back to the small one.

4. Each pair of triangles is similar. Find the length marked x.

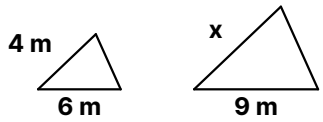
a)



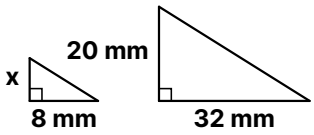
b)



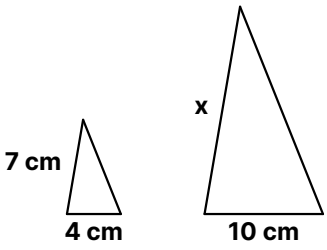
c)



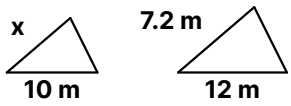
d)



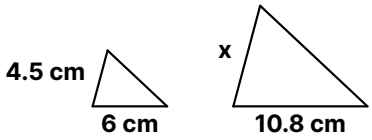
e)



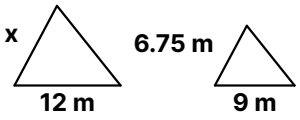
f)



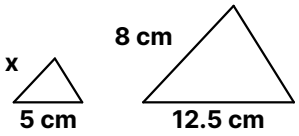
g)



h)



i)

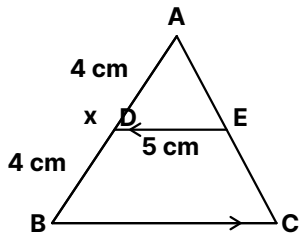


D A triangle inside a triangle ●●○

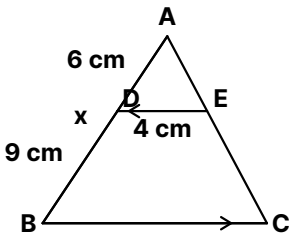
Remember: A line parallel to one side cuts off a small triangle that is similar to the whole triangle, because the angles are the same (corresponding angles). Use the whole side, not the piece.

5. In each diagram DE is parallel to BC. Find x.

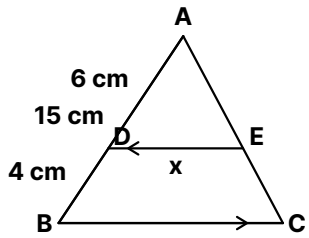
a)



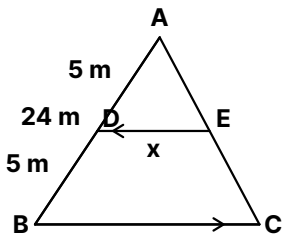
b)



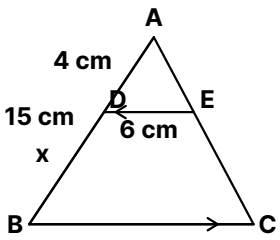
c)



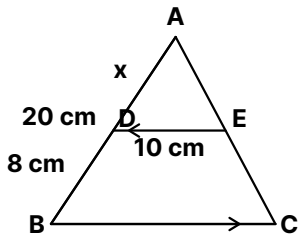
d)



e)



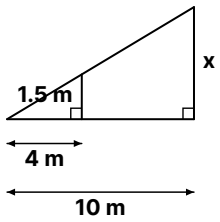
f)



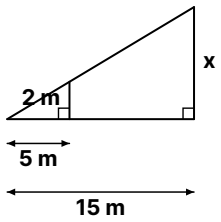
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6. The small right-angled triangle shares a corner with the large one. Find the height marked x.

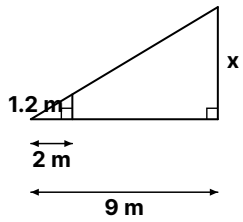
a)



b)



c)

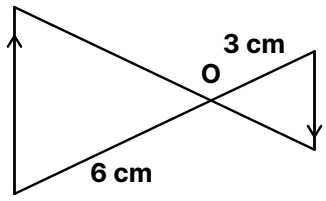


E Negative scale factors ●●●

> ⚠ **Watch out:** When the image is on the other side of the centre O, the triangles point in opposite directions and the scale factor is negative. The lengths still scale by the size of k. The two triangles are similar because the angles at O are vertically opposite and the parallel sides give alternate angles.

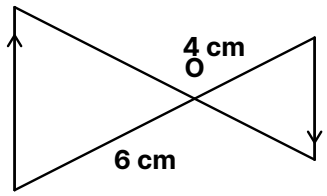
7. Each large triangle is an enlargement of the small one with centre O. Find the scale factor k, with its sign.

a)



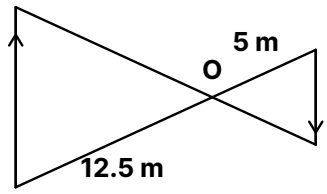
$k =$

b)



$k =$

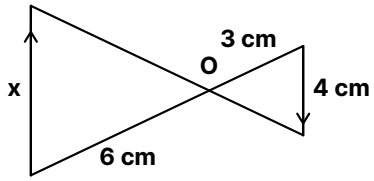
c)



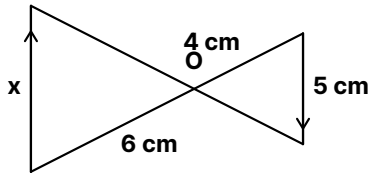
$k =$

8. The triangles share the point O and the marked sides are parallel. Find x.

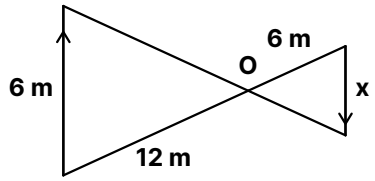
a)



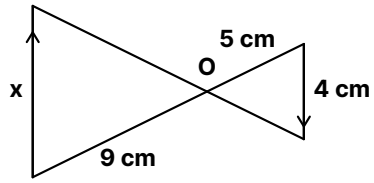
b)



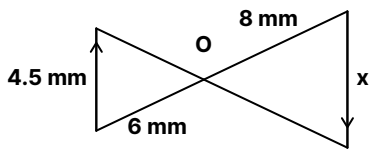
c)



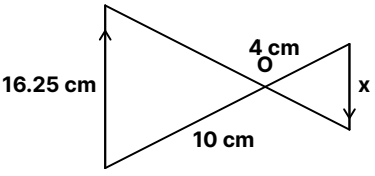
d)



e)



f)

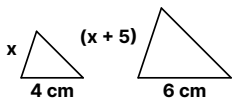


F Sides in terms of x ●●●

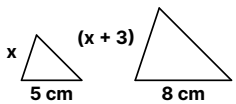
> 💡 **Remember:** Write the two matching sides as a fraction equal to the scale factor, then solve the equation. Check that your x gives sides in the right ratio.

9. The pairs of triangles are similar, with the matching sides labelled. Find x.

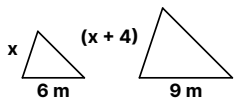
a)



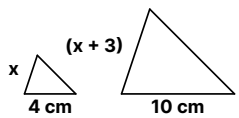
b)



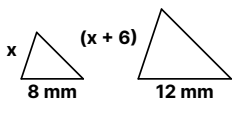
c)



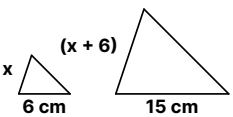
d)



e)



f)



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G Problems ●●●

10. On a map a triangular park has sides of 4 cm, 5 cm and 6 cm. The shortest side of the real park is 120 m. Find the perimeter of the real park.

11. A person 1.8 m tall has a shadow 2.4 m long. At the same time a tree has a shadow 12 m long. How tall is the tree?

12. A photo is 10 cm wide and 15 cm long. It is enlarged so that it is 24 cm wide. How long is the enlargement?

13. A 2 m post has a shadow 1.5 m long. A flagpole beside it has a shadow 7.5 m long. How tall is the flagpole?

14. To find the width of a river, a surveyor marks two similar triangles: one with sides 12 m and 8 m on the near bank, and a larger one whose matching 8 m side is 30 m. How wide is the river (the side matching the 12 m side)?

15. A triangle on a map has sides of 3 cm, 4 cm and 5 cm. The shortest side of the real triangle is 600 m. Find the other two sides of the real triangle.

★ Extension

16. Two similar triangles have areas of 20 cm² and 180 cm². Find the scale factor from the small one to the large one.

17. Two triangles are similar. Sides of x cm and 6 cm on the small triangle match sides of $(x + 4)$ cm and 9 cm on the large one. Find x.

18. A triangle has sides of 5 cm, 6 cm and 7 cm. A similar triangle is drawn using a scale factor of 2.5. Find the perimeter of the new triangle.

19. A triangle is enlarged by scale factor 3. Its area was 12 cm². What is the area of the enlargement? (Think about what happens to both the base and the height.)

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Similar Triangles

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

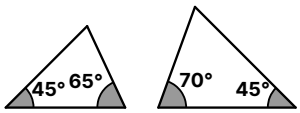
KEY WORDS similar *(rite)* · scale factor *(tauwehe tauine)* · corresponding sides · enlargement *(whakarahi)*

A Are they similar? ●○○

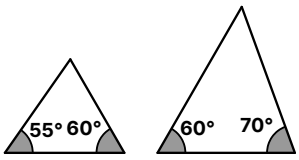
Remember: Two triangles are similar when they have the same three angles. Two matching angles are enough: the third angle is then the same too.

1. Decide whether each pair of triangles is similar. Give a reason.

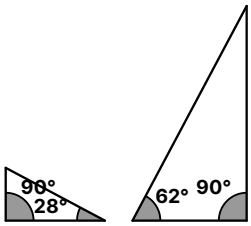
a)



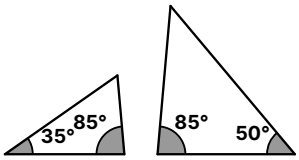
b)



c)



d)



2. These are the side lengths of two triangles. Are the triangles similar?

a) 5, 12, 13 cm and 12.5, 30, 32.5 cm

b) 4, 5, 6 cm and 10, 12.5, 16 cm

c) 6, 9, 12 mm and 8, 12, 16 mm

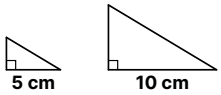
d) 3, 5, 7 m and 9, 14, 21 m

B Scale factor ●○○

Remember: Scale factor k = a side of the image ÷ the matching side of the original. Match the sides that are in the same position.

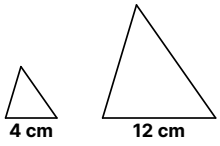
3. Each pair of triangles is similar. Find the scale factor k from the small triangle to the large one.

a)



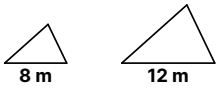
$k =$ _____

b)



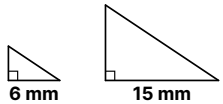
$k =$ _____

c)



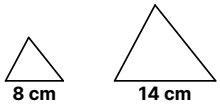
$k =$ _____

d)



$k =$ _____

e)



$k =$ _____

f)



$k =$ _____

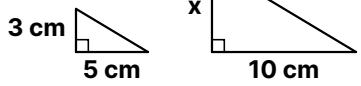
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C Finding a missing length ●●○

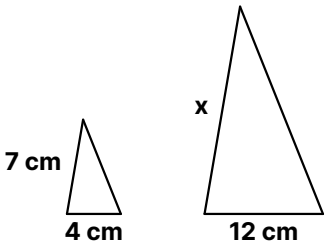
Remember: Find k from a pair of sides you know. Multiply by k to go to the large triangle, divide by k to come back to the small one.

4. Each pair of triangles is similar. Find the length marked x.

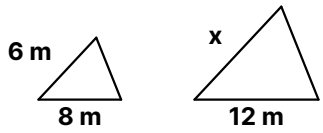
a)



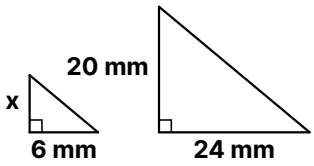
b)



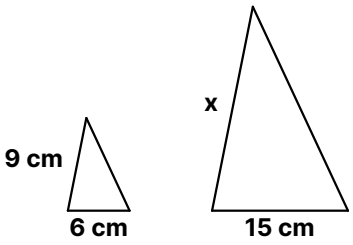
c)



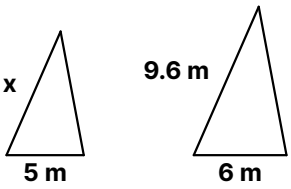
d)



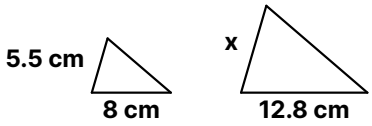
e)



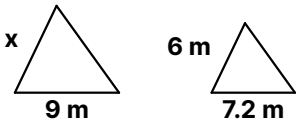
f)



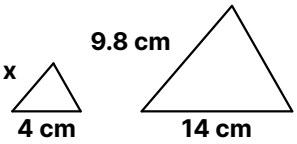
g)



h)



i)

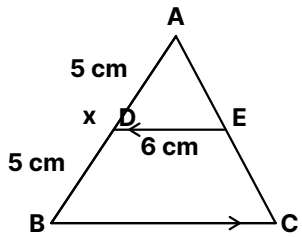


D A triangle inside a triangle ●●○

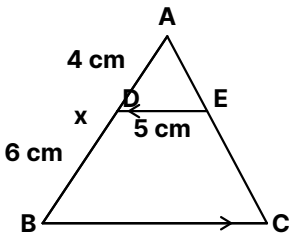
Remember: A line parallel to one side cuts off a small triangle that is similar to the whole triangle, because the angles are the same (corresponding angles). Use the whole side, not the piece.

5. In each diagram DE is parallel to BC. Find x.

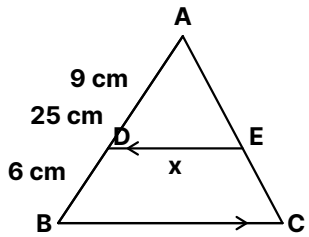
a)



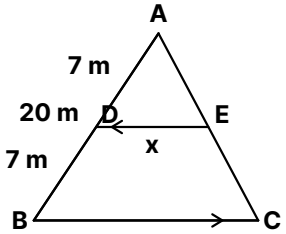
b)



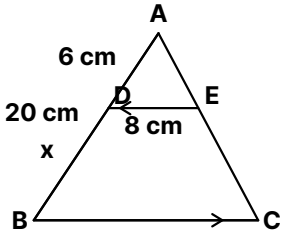
c)



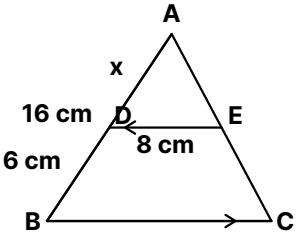
d)



e)



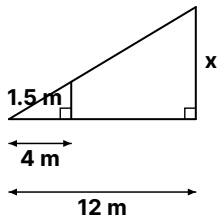
f)



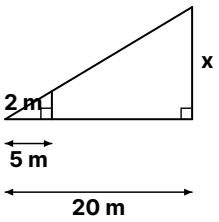
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6. The small right-angled triangle shares a corner with the large one. Find the height marked x.

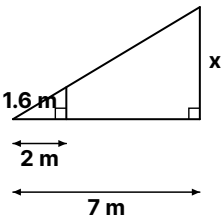
a)



b)



c)

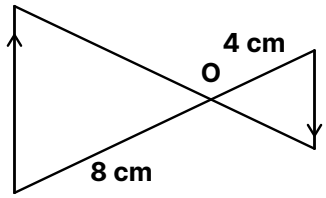


E Negative scale factors ●●●

> ⚠ **Watch out:** When the image is on the other side of the centre O, the triangles point in opposite directions and the scale factor is negative. The lengths still scale by the size of k. The two triangles are similar because the angles at O are vertically opposite and the parallel sides give alternate angles.

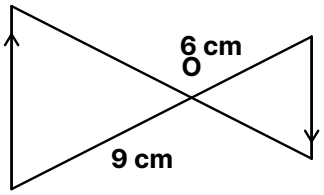
7. Each large triangle is an enlargement of the small one with centre O. Find the scale factor k, with its sign.

a)



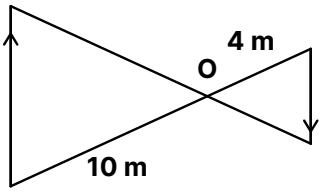
$k =$

b)



$k =$

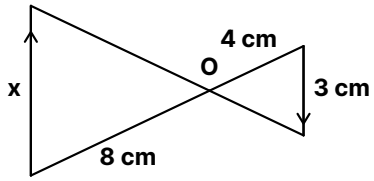
c)



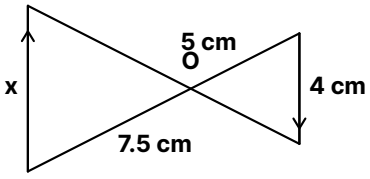
$k =$

8. The triangles share the point O and the marked sides are parallel. Find x.

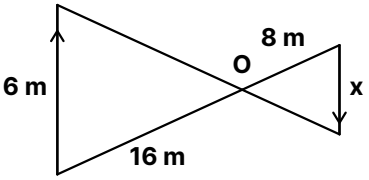
a)



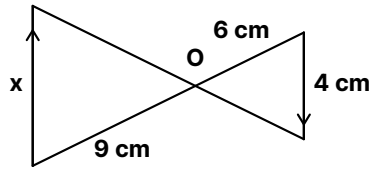
b)



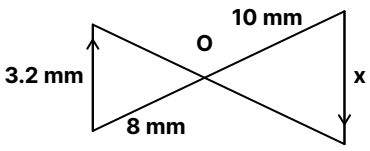
c)



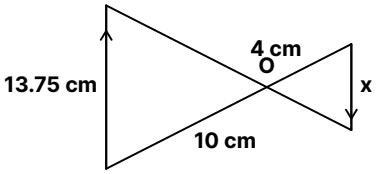
d)



e)



f)

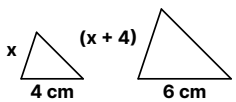


F Sides in terms of x ●●●

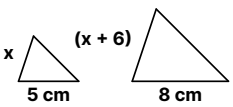
> 💡 **Remember:** Write the two matching sides as a fraction equal to the scale factor, then solve the equation. Check that your x gives sides in the right ratio.

9. The pairs of triangles are similar, with the matching sides labelled. Find x.

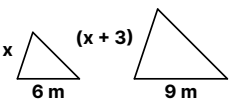
a)



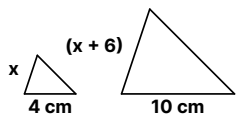
b)



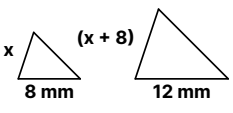
c)



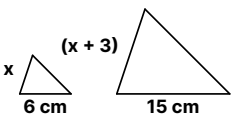
d)



e)



f)



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G Problems ●●●

10. On a map a triangular park has sides of 3 cm, 5 cm and 7 cm. The shortest side of the real park is 150 m. Find the perimeter of the real park.

11. A person 1.6 m tall has a shadow 2 m long. At the same time a tree has a shadow 15 m long. How tall is the tree?

12. A photo is 12 cm wide and 18 cm long. It is enlarged so that it is 30 cm wide. How long is the enlargement?

13. A 3 m post has a shadow 2 m long. A flagpole beside it has a shadow 9 m long. How tall is the flagpole?

14. To find the width of a river, a surveyor marks two similar triangles: one with sides 15 m and 9 m on the near bank, and a larger one whose matching 9 m side is 40 m. How wide is the river (the side matching the 15 m side)?

15. A triangle on a map has sides of 5 cm, 12 cm and 13 cm. The shortest side of the real triangle is 250 m. Find the other two sides of the real triangle.

★ Extension

16. Two similar triangles have areas of 15 cm² and 240 cm². Find the scale factor from the small one to the large one.

17. Two triangles are similar. Sides of x cm and 4 cm on the small triangle match sides of $(x + 6)$ cm and 10 cm on the large one. Find x.

18. A triangle has sides of 4 cm, 7 cm and 9 cm. A similar triangle is drawn using a scale factor of 3.5. Find the perimeter of the new triangle.

19. A triangle is enlarged by scale factor 2.5. Its area was 20 cm². What is the area of the enlargement? (Think about what happens to both the base and the height.)

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